

Chapter 9

Slice Category

9.1 Definition

Definition 9.1.1 (Slice Category). For any category $\mathcal{C}$ and $\mathcal{C}$ -object A , there is a *slice category* $\mathcal{C}/A$. The objects of $\mathcal{C}/A$ are $\mathcal{C}$ -arrows with codomain A . If there exist $\mathcal{C}$ -arrows $f : B \rightarrow A$, $g : C \rightarrow A$, $h : B \rightarrow C$ and the following triangle commutes, then h is an $\mathcal{C}/A$ -arrow with domain f and codomain g . Composition and identity arrows in $\mathcal{C}/A$ are those of A .

$$\begin{array}{ccc} B & \xrightarrow{h} & C \\ & \searrow f \quad \swarrow g & \\ & A & \end{array}$$

Exercise 9.1.2. Show that if $\mathcal{C}$ has a terminal object 1 , then $\mathcal{C}/1$ is equivalent to $\mathcal{C}$.

Exercise 9.1.3. Show that, for any category $\mathcal{C}$, each slice $\mathcal{C}/A$ has a terminal object, whether or not $\mathcal{C}$ has a terminal object.

Theorem 9.1.4. If $\mathcal{C}$ has all finite limits, then so does each slice $\mathcal{C}/A$.

Proof. The following pullback is the binary product for $\mathcal{C}/B$ -objects $f : A \rightarrow B$ and $g : C \rightarrow B$:

$$\begin{array}{ccc} A \times_B C & \xrightarrow{p_2} & C \\ p_1 \downarrow & & \downarrow g \\ A & \xrightarrow{f} & B \end{array}$$

For any $\mathcal{C}/B$ -arrows $k : f \rightarrow g$ and $k' : f \rightarrow g$, the following diagram is an equalizer:

$$\begin{array}{ccccc} E & \xrightarrow{e} & A & \xrightleftharpoons[k]{k'} & C \\ & \searrow & \downarrow f & \swarrow g & \\ & f \circ e & B & & \end{array}$$

where E, e is an equalizer for k and k' in $\mathcal{C}$. □

9.2 Slice Functors

Definition 9.2.1 (Functor A^*). If a category $\mathcal{C}$ has binary products, there is a functor $A^* : \mathcal{C} \rightarrow \mathcal{C}/A$ for any $\mathcal{C}$ -object A , which takes any $\mathcal{C}$ -object B to $\pi_1 : A \times B \rightarrow A$, and any $\mathcal{C}$ -arrow $f : B \rightarrow C$ to $id_A \times f$.

$$\begin{array}{ccc} B & \xrightarrow{f} & C \\ & & \uparrow \pi_1 \\ A \times B & \xrightarrow{id_A \times f} & A \times C \\ & \searrow \pi_1 & \swarrow \pi_1 \\ & A & \end{array}$$

Definition 9.2.2 (Functor Σ_A). For a category $\mathcal{C}$, there is a functor $\Sigma_A : \mathcal{C}/A \rightarrow \mathcal{C}/A$ for any $\mathcal{C}$ -object A , which takes any $\mathcal{C}/A$ object $f : B \rightarrow A$ to B , and any $\mathcal{C}/A$ arrow h to itself.

$$\begin{array}{ccc} B & \xrightarrow{h} & C \\ & \searrow f & \swarrow g \\ & A & \end{array} \qquad B \xrightarrow{h} C$$

Theorem 9.2.3 ($\Sigma_A \dashv A^*$). If $\mathcal{C}$ has binary products, then for every $\mathcal{C}$ object A , the functor Σ_A is left adjoint to A^* .

Proof. Consider the following diagram between $\mathcal{C}/A$ and $\mathcal{C}$, where A, B are $\mathcal{C}$ -objects and $g : C \rightarrow A$ is a $\mathcal{C}/A$ object. To show Theorem 9.2.3, it suffices to construct a universal arrow (A^*B, ε_g) from Σ_A to B for all $\mathcal{C}/A$ -object g , such that for any $f : \Sigma_A g \rightarrow B$, there exists a unique adjunct $\bar{f} : g \rightarrow A^*B$ that makes the following diagram commute:

$$\begin{array}{ccc}
 A^*B & & \Sigma_A A^*B \xrightarrow{\varepsilon_g} B \\
 \bar{f} \uparrow & & \Sigma \bar{f} \uparrow \searrow f \\
 g & & \Sigma_A g
 \end{array}$$

Since $\Sigma_A A^*B = \Sigma_A \pi_1 = A \times B$ and $\Sigma_A g = C$, the above diagram amounts to the following diagram:

$$\begin{array}{ccc}
 \pi_1 & & A \times B \xrightarrow{\varepsilon_g} B \\
 \bar{f} \uparrow & & \Sigma \bar{f} \uparrow \searrow f \\
 g & & C
 \end{array}$$

where the left diagram in $\mathcal{C}/A$ is the following triangle, which means that $\bar{f}$ must be of the form $\langle g, h \rangle$ for some $h : C \rightarrow B$.

$$\begin{array}{ccc}
 C & \xrightarrow{\bar{f}} & A \times B \\
 g \searrow & & \swarrow \pi_1 \\
 & A &
 \end{array}$$

Let us define the adjunct of f as $\langle g, f \rangle$, which is completely determined by f . Then $\Sigma \bar{f} = \Sigma \langle g, f \rangle = \langle g, f \rangle$. Thus let us further define ε_g as π_2 , which makes (π_1, π_2) a universal arrow from Σ_A to B . The uniqueness of the adjunct is obvious.

$$\begin{array}{ccc}
 \pi_1 & & A \times B \xrightarrow{\pi_2} B \\
 \langle g, f \rangle \uparrow & & \langle g, f \rangle \uparrow \searrow f \\
 g & & C
 \end{array}$$

By Theorem 6.2.7, η_g is the adjunct of $id_{\Sigma_A g} = id_C$, which is $\langle g, id_C \rangle$. $\square$

Definition 9.2.4 (Internal products). If $\mathcal{C}$ is Cartesian closed, any A -indexed family of objects $f : B \rightarrow A$ has an A -indexed product $\Pi_A f$ defined by the following pullback.

$$\begin{array}{ccc} \Pi_A f & \longrightarrow & 1 \\ \downarrow & & \downarrow [id_A] \\ B^A & \xrightarrow{f^A} & A^A \end{array}$$

where $[id_A]$ (called the *name of id_A*) is defined as the transpose of $\pi_2 : 1 \times A \rightarrow A$.

Remark 9.2.5. Intuition behind Definition 9.2.4 is that Π_A is the subset of B^A consisting of internal right-inverses of f . In other words, $\Pi_A f$ is a collection of a A -tuple each of whose element is chosen from each fibre introduced by f .

Definition 9.2.6 (Functor Π_A). If $\mathcal{C}$ is Cartesian closed, there is a functor $\Pi_A : \mathcal{C}/A \rightarrow \mathcal{C}$ for any $\mathcal{C}$ -object A , which takes each $\mathcal{C}/A$ -object $f : B \rightarrow A$ to $\Pi_A f$, and each $\mathcal{C}/A$ -arrow $k : f \rightarrow g$ to $\Pi_A k : \Pi_A f \rightarrow \Pi_A g$ by pulling the following triangle back along $[id_A]$.

$$\begin{array}{ccc} B^A & \xrightarrow{k^A} & C^A \\ & \searrow f^A \quad \swarrow g^A & \\ & A^A & \end{array}$$

Remark 9.2.7. The existence of $\Pi_A k$ is ensured by Exercise 2.3.6.

$$\begin{array}{ccccc} & & \Pi_A g & \xrightarrow{\quad} & C^A \\ & \nearrow \Pi_A k & & \nearrow k^A & \\ \Pi_A f & \xrightarrow{\quad} & B^A & \xrightarrow{\quad} & A^A \\ & \searrow & \searrow f^A & \searrow g^A & \\ & 1 & \xrightarrow{[id_A]} & & \end{array}$$

Theorem 9.2.8 ($A^* \dashv \Pi_A$). The functor Π_A is right adjoint to A^* .

Proof. It suffices to show that there is a functor isomorphism between the comma categories $(A^*, \mathcal{C}/A)$ and $(\mathcal{C}, \Pi_A)$.

Let $f : B \rightarrow A$ be a $\mathcal{C}$ -arrow. Consider the following series of isomorphisms:

$$\begin{array}{c}
 \frac{T \xrightarrow{t} \Pi_A f}{T \xrightarrow{p_1 \circ t} B^A} \quad \text{in } \mathcal{C} \\
 \hline
 \frac{T \times A \xrightarrow{\overline{p_1 \circ t}} B}{A \times T \xrightarrow{\sim} T \times A \xrightarrow{\overline{p_1 \circ t}} B} \quad \text{such that } f \circ \overline{p_1} \circ t = \pi_2 \\
 \hline
 \frac{A \times T \xrightarrow{\sim} T \times A \xrightarrow{\overline{p_1 \circ t}} B}{A^* T \xrightarrow{(\overline{p_1 \circ t}) \circ \sim} f} \quad \text{such that } f \circ \overline{p_1} \circ t = \pi_1 \\
 \hline
 A^* T \xrightarrow{(\overline{p_1 \circ t}) \circ \sim} f \quad \text{in } \mathcal{C}/A
 \end{array}$$

Note that the arrow $\overline{p_1} \circ t \circ \sim$ is uniquely determined by t . In the third step, we need $f \circ \overline{p_1} \circ t \circ \sim = \pi_1$ to regard it as $\mathcal{C}/A$ arrow. This holds since:

$$\begin{aligned}
 & f \circ (\overline{p_1} \circ t) \circ \sim \\
 = & ev \circ (f^A \times id_A) \circ (p_1 \circ t \times id_A) \circ \sim \\
 = & ev \circ ((f^A \circ p_1 \circ t) \times id_A) \circ \sim \\
 = & ev \circ ([id_A] \circ !_T \times id_A) \circ \sim \\
 = & ev \circ ([id_A] \times id_A) \circ (!_T \times id_A) \circ \sim \\
 = & \pi_2 \circ (!_T \times id_A) \circ \sim \\
 = & id_A \circ \pi_2 \circ \sim \\
 = & \pi_1
 \end{aligned}$$

$$\begin{array}{ccccc}
 A^A & & A^A \times A & \xrightarrow{ev} & A \\
 f^A \uparrow & & f^A \times id_A \uparrow & & \uparrow f \\
 B^A & & B^A \times A & \xrightarrow{ev} & B \\
 \pi_1 \circ t \uparrow & & (p_1 \circ t) \times id_A \uparrow & \nearrow \overline{p_1 \circ t} & \\
 T & & T \times A & & \\
 & & \sim \uparrow & & \\
 & & A \times T & &
 \end{array}$$

Then we need to show that the right square in $\mathcal{C}/A$ commutes if the left square in $\mathcal{C}$ commutes. This is left as an exercise.

$$\begin{array}{ccccc}
 T & \xrightarrow{t} & \Pi_A f & & f & T & \xrightarrow{\quad} & f \\
 h \downarrow & & \Pi_A k \downarrow & & \downarrow k & h \downarrow & & \downarrow k \\
 T' & \xrightarrow{t'} & \Pi_A g & & g & T' & \xrightarrow{\quad} & g \\
 & & & & & A^* h \downarrow & & A^* T' \xrightarrow{\quad} g
 \end{array}$$

The above series of arrows could be traced from the bottom to the top:

$$\begin{array}{c}
\frac{A^*T \xrightarrow{r} f}{A \times T \xrightarrow{r} B} \quad \text{in } \mathcal{C}/A \\
\hline
\frac{T \times A \xrightarrow{\sim} A \times T \xrightarrow{r} B}{T \xrightarrow{\lambda(r \circ \sim)} B^A} \\
\hline
T \xrightarrow{!} \Pi_A f \quad (\Pi_A f \text{ is a pullback})
\end{array}$$

Again, the arrow in the last line is uniquely determined by r . By using this transition, we can show that the right square in $\mathcal{C}$ commutes if the left square in $\mathcal{C}/A$ commutes. This is also left as an exercise.

$$\begin{array}{ccccc}
T & A^*T & \xrightarrow{r} & f & T & \longrightarrow & \Pi_A f & f \\
h \downarrow & A^*h \downarrow & & \downarrow k & h \downarrow & & \Pi_A k \downarrow & \downarrow k \\
T' & A^*T' & \xrightarrow{r'} & g & T' & \longrightarrow & \Pi_A g & g
\end{array} \quad \square$$

Exercise 9.2.9. Complete the proof of Theorem 9.2.8.

Theorem 9.2.10. Suppose that a category $\mathcal{C}$ has all finite limits. For each $\mathcal{C}$ -object A , if the slice functor A^* has a right adjoint $\Pi_A : \mathcal{C}/A \rightarrow \mathcal{C}$, then $\mathcal{C}$ is a Cartesian closed.

Proof. $_ \times A$ factors as $\Sigma_A \circ A^*$. But $\Sigma_A \dashv A^*$ (by Theorem 9.2.3) and $A^* \dashv \Pi_A$ (by Theorem 9.2.8). By Theorem 6.5.1, $\Sigma_A \circ A^* \dashv \Pi_A \circ A^*$, which gives rise to the exponential. $\square$

Theorem 9.2.11. For any category with binary products, suppose that B^C is an exponential of B by C . Then, for each $\mathcal{A}$ -object A , $A^*(B^C)$ is an exponential of A^*B by A^*C in $\mathcal{C}/A$.

Proof. Let $g : C \rightarrow A$ be a $\mathcal{C}$ -arrow. For any $\mathcal{C}/A$ -arrow $f : g \times_A A^*C \rightarrow A^*B$, consider the following series of arrows:

$$\begin{array}{c}
\frac{g \times_A A^*C \xrightarrow{f} A^*B}{\Sigma_A(g \times_A A^*C) \rightarrow B} \quad \text{in } \mathcal{C}/A \\
\hline
\frac{(\Sigma_A g) \times C \rightarrow B}{\Sigma_A g \rightarrow B^C} \quad (\text{By Exercise 2.3.3}) \\
\hline
g \rightarrow A^*(B^C) \quad \text{in } \mathcal{C}/A
\end{array}$$

Let λf be the $\mathcal{C}/A$ -arrow in the last line. Then, it suffices to show that the following square commutes:

$$\begin{array}{ccc}
 A^*(B^C) & A^*(B^C) \times_A A^*C & \xrightarrow{ev} A^*B \\
 \lambda f \uparrow & \lambda f \times_A id_{A^*C} \uparrow & \nearrow f \\
 g & g \times_A A^*C &
 \end{array}$$

□

Exercise 9.2.12. Complete the proof of Theorem 9.2.11.

9.3 Functors between Slices

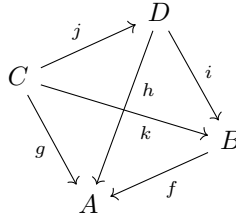
Remark 9.3.1. Let $\mathcal{C}$ be a category and $f : B \rightarrow A$ is a $\mathcal{C}$ -arrow. There is a functor $\Sigma_f : \mathcal{C}/B \rightarrow \mathcal{C}/A$ that sends each $\mathcal{C}/B$ -object $g : C \rightarrow B$ to $f \circ g : C \rightarrow A$, and each $\mathcal{C}/B$ -arrow $k : g \rightarrow h$ to $k : f \circ g \rightarrow f \circ h$.

We use the same symbol Σ_f for this functor as the functor $\Sigma : \mathcal{C}/A/f \rightarrow \mathcal{C}/A$ but this ambiguity is tolerated by the following isomorphism:

Theorem 9.3.2. For any $\mathcal{C}$ -arrow $f : B \rightarrow A$ there exists a functor isomorphism I that makes the following triangle commute:

$$\begin{array}{ccc}
 (\mathcal{C}/A)/f & \xrightarrow{I} & \mathcal{C}/B \\
 \searrow \Sigma_f & & \swarrow \Sigma_f \\
 & \mathcal{C}/A &
 \end{array}$$

Proof. Each $(\mathcal{C}/A)/f$ -object is a $\mathcal{C}/A$ arrow with codomain f (ex. $k : g \rightarrow f$, $i : h \rightarrow f$), which is also a $\mathcal{C}/B$ object (ex. $k : C \rightarrow B$, $i : D \rightarrow B$). Each $\mathcal{C}/B$ -object $x : X \rightarrow B$ is a $\mathcal{C}$ -arrow with codomain B , which sends $\mathcal{C}$ -arrow $f \circ x : X \rightarrow A$ to $f : B \rightarrow A$, thus is a $\mathcal{C}/A$ arrow with codomain f .



It is left as an exercise to show that each $(\mathcal{C}/A)/f$ -arrow is a $\mathcal{C}/A$ -arrow, and each $\mathcal{C}/A$ -arrow is a $(\mathcal{C}/A)/f$ -arrow. $\square$

Exercise 9.3.3. Complete the proof of Theorem 9.3.2.

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